Name:______________________________ Period:________

Advanced:
Solving Systems of Equations

1. $-8x - 9y - 10z = -95$
$9\,(2x + 2y + z = -84)10$
$-10x - 9y - 9z = 11$

$-10x - 9y - \cancel{9z} = 11$
$18x + 18y + \cancel{9z} = -756$
$8x + 9y = -745$

$8x + 9(-73) = -745$
$8x - 657 = -745$
$\frac{8x}{8} = \frac{-88}{8}$
$x = -11$
$\boxed{(-11, -73, 84)}$

$-8x - 9y - 10z = -95$
$20x + 20y + \cancel{10z} = -840$
$(12x + 11y = -935)2$
$(8x + 9y = -745)3$

$24x + 22y = -1870$
$-24x - 27y = 2235$
$\frac{-5y}{-5} = \frac{365}{-5}$
$y = -73$

$2(-11) + 2(-73) + z = -84$
$-168 + z = -84$
$z = 84$

2. $-x + 3y + 2z = -2$
$-3x - 3y + 2z = 2$
$-x - 3y + z = 9$

$-x + \cancel{3y} + 2z = -2$
$-x - \cancel{3y} + z = 9$
$-2x + 3z = 7$

$-x + \cancel{3y} + 2z = -2$
$-3x - \cancel{3y} + 2z = 2$
$-4x + 4z = 0$
$-x + z = 0$
$\cancel{2x} - 2z = 0$
$-\cancel{2x} + 3z = 7$
$z = 7$

$-x + 7 = 0$
$7 = x$

$-7 + 3y + 2(7) = -2$
$-7 + 3y + 14 = -2$
$3y = -9$
$y = -3$

$(7, -3, 7)$